Crypto Range Screener

Introduction

This project analyzes cryptocurrency price movement using three trading signals: Range, Sweep, and Failure. These signals are detected from historical OHLCV data. Based on these signals, the program creates trade entries automatically using fixed rules. A Random Forest machine learning model is then used to predict future price movement. Stablecoin market data from DefiLlama is added as an on-chain feature to check whether blockchain data improves the prediction.

1. Signal Behavior Prediction

Range Signal - The price is moving inside a small range without a clear trend. Buyers and sellers are balanced, so the price moves between the range high and range low. I do not open a trade immediately after a Range Signal. Instead, I wait for either a Sweep Signal or a Failure Signal.

Market Logic - As the price stays inside the range, buy and sell orders build near support and resistance. After enough liquidity is collected, the market usually makes a stronger move.

Sweep Signal – This happens when the price moves outside the range but closes back inside. This usually means stop-loss orders outside the range have been triggered.

Market Logic - After collecting this liquidity, the breakout loses strength and the price often reverses toward the opposite side of the range.

Failure Signal - This happens when the price closes outside the range but returns inside within six candles. This shows the breakout was not strong enough to continue.

Market Logic - When the breakout fails, traders close their positions, which often pushes the price back toward the opposite side of the range.

2. Entry Strategy

Range: No trade is taken. I wait for a Sweep or Failure Signal.

Sweep: Sell when price moves above the range and closes back inside. Buy when price moves below the range and closes back inside. Entry is the closing price of the sweep candle. Stop-loss is just above the high for Sell or just below the low for Buy. Target is the opposite side of the range.

Failure: Sell when an upward breakout fails and returns inside the range. Buy when a downward breakout fails and returns inside. Entry is the closing price of the failure candle. Stop-loss is placed just outside the candle. Target is the opposite side of the range.

3. Why I Chose This Strategy

The strategy is completely rule-based. Every signal, entry, stop-loss, and target is generated using fixed conditions in the program. This removes manual decisions, makes the strategy easy to test on historical data, and produces the same result every time the program runs.

4. Machine Learning Model

I chose the Random Forest Regressor because cryptocurrency prices are influenced by many factors and do not always follow a simple pattern. Random Forest can learn complex relationships between different features and works well with both numerical values and signal-based features. It also does not require feature scaling, making it simple and reliable for this project.

The model uses these features:

- **6-hour return:** Captures the short-term trend.
- **1-hour return:** Shows recent price movement.
- **Range width:** Shows whether the market is consolidating.
- **Range Signal:** Identifies consolidation.
- **Sweep Signal:** Detects liquidity grabs.
- **Failure Signal:** Detects failed breakouts.
- **Stablecoin flow:** Adds market liquidity information.

A second model also includes stablecoin flow from DefiLlama.

The target is the future 6-hour price return. The data is split into 80% training and 20% testing without shuffling because the data follows a time order. The models are evaluated using Mean Absolute Error (MAE) and Directional Accuracy. The program also plots predicted values against actual values to visually compare the forecasts.

5. Blockchain / Web3 Analytics

I used stablecoin market capitalization data from the DefiLlama API because it is a free and publicly available source. Stablecoins are widely used to buy cryptocurrencies, so changes in stablecoin supply can give useful information about money entering or leaving the market.

The program calculates the percentage change in stablecoin market capitalization and uses it as a stablecoin flow feature.

Two Random Forest models are trained. The first uses only price-based features, while the second includes stablecoin flow. Both models are tested using the same data, and their MAE and Directional Accuracy are compared.

I chose this approach because it allows me to check whether adding blockchain data improves the prediction. The on-chain data is not used to replace technical signals; it is used to provide extra market information that may improve the model's performance.

Conclusion

This project combines technical analysis, machine learning, and blockchain analytics in a simple rule-based system. The strategy is easy to understand, produces consistent results, and allows a clear comparison between predictions made with and without on-chain data.